Theorem proving

Direct proof · contrapositive · contradiction · disproof by counterexample

Today's goal: turn a mathematical statement into a clear argument that shows **why it must be true in every case it claims**.

A proof is not a list of examples. It is a chain of valid deductions from stated assumptions, definitions, and previously known facts.

Session	Date
5 of 15	Fri 11 Sep 2026

COURSE OUTLINE

Where session 5 sits

1 · Course overview and school-maths refresher	2 · Sets and set operations
3 · Functions	4 · Function families
5 · Theorem proving	6 · Mathematical induction
7 · Elementary combinatorics and binomial theorem	8 · Midterm
9 · GCD and divisibility	10 · Modular arithmetic
11 · Vectors	12 · Polynomials
13 · Complex numbers	14 · Course recap and exam preparation
15 · Final exam	

TODAY Direct proof, contrapositive, contradiction, and counterexamples — the tools used to justify the precise claims introduced in earlier sessions.

Function families, in four ideas

TRANSFORMATIONS · SINE Inside moves the input; outside moves the output. In $A\sin\big(B(t-h)\big)+D$ with $B>0$: amplitude $|A|$, period $2\pi/B$, midline $y=D$.

POLYNOMIALS The leading term decides the ends; roots give factors. $p(r)=0 \iff (x-r)$ is a factor, so degree n means at most n real roots.

EXPONENTIALS & LOGS A log is an exponent: $\log_b x = y \iff b^y = x$, so its input must be positive. $\log(uv)=\log u+\log v$ only if $u,v>0$; $\log(u+v)$ does not split in general.

GROWTH "Eventually" means from some point on. $\log_2 n < n^2 < 2^n$ for every $n\geq 5$, not for every n: at $n=2$, $n^2=2^n$.

WARM-UP One pair disproves the $\log_2(u+v)$ rule. Why can't examples prove the $\log_2(uv)$ law?

A theorem is more than a pattern

DEFINITION A **mathematical statement** is a declarative sentence with a definite truth value: once variables, domains and quantifiers are fixed, it is true or false.

DEFINITION A **theorem** is such a statement established by proof from definitions, axioms and earlier results.

SENTENCE	STATEMENT?	WHY?
"$x + 1 = 4$"	No	It is an **open sentence**: its truth depends on the unspecified value of x.
"For every real x, $x^2 \geq 0$"	Yes	It makes one claim about all real numbers.
"There is an integer n with $n^2 = 9$"	Yes	It says that at least one integer works.

QUANTIFIERS "For every" means **all allowed values**; "there is" means **at least one allowed value**.

An implication has a hypothesis and a conclusion

DEFINITION An implication has the form

$$P \implies Q,$$

read as "if P, then Q."

- P is the hypothesis (or condition).
- Q is the conclusion.

EXAMPLE · WITH ITS DOMAIN For every $n \in \mathbb{Z}$, if n is a multiple of 4, then n is even.

THE WORDS, IN ALGEBRA $\mathbb{Z}$ is the set of integers. "Multiple of 4" means $n = 4k$ for some $k \in \mathbb{Z}$; "even" means $n = 2m$ for some $m \in \mathbb{Z}$.

ONE DIRECTION ONLY An implication only tells us what follows when its hypothesis holds. It does not say that every even integer is a multiple of 4.

An implication with no cases to check is true

VACUOUS TRUTH

$P \Rightarrow Q$ claims there is **no case** where P holds and Q fails.

If P never holds, there is nothing to check — the implication is true *vacuously*.

EXAMPLE "Every integer n with $n^2 < 0$ is prime" is **true**. No such n exists.

WHY IT MATTERS A vacuously true statement tells you nothing. If your proof works only because the hypothesis is never satisfied, you have proved nothing useful.

"Without loss of generality" must be earned

WLOG

Write WLOG when a symmetry lets you assume one of several cases, because the others follow by relabelling.

It shortens a proof. It never skips one.

LEGITIMATE To prove $|a+b| \leq |a| + |b|$, say "WLOG $|a| \geq |b|$" — swapping the names a and b changes neither side.

THE TEST You must be able to name the relabelling. "WLOG a is even" is illegal unless something genuinely makes the odd case identical.

Do not confuse an implication with its converse

NOTATION Fix one domain and consider an implication $P \Rightarrow Q$ in that domain. Here, $\neg P$ means "not P."

NAME	FORM
original statement	$P \Rightarrow Q$
converse	$Q \Rightarrow P$
contrapositive	$\neg Q \Rightarrow \neg P$

THEOREM The original statement and its contrapositive are logically equivalent: over the same domain, they are always both true or both false. The converse is a different statement and may have a different truth value.

EXAMPLE "multiple of $4 \Rightarrow$ even" is true. Its converse, "even $\Rightarrow$ multiple of 4," is false: 2 is even and is not a multiple of 4.

ACTIVITY 1

Read the logical direction

ACTIVITY 1 PAIRS 10 MIN

Let $P(n)$ mean "n is even" and $Q(n)$ mean "n^2 is even," where $n \in \mathbb{Z}$.

1. Write the converse of $P(n) \Rightarrow Q(n)$ in words.
2. Write the contrapositive of $P(n) \Rightarrow Q(n)$ in words.
3. For "if n is a multiple of 4, then n is even," test its converse over $\mathbb{Z}$. Is it true or false? Justify your answer.

Share: one pair reads their converse aloud; the room decides whether it is true, and names the witness if not.

Converse: swap. Contrapositive: swap and negate.

METHOD For $P \Rightarrow Q$:

1. **Converse:** swap the two parts: $Q \Rightarrow P$.
2. **Contrapositive:** swap **and** negate both: eg Q\Rightarrow\neg P1

PARITY OF INTEGERS For \in\mathbb Z " odd" means =2k+11 Every integer is even or odd, never both — so "not even" means "odd".

REQUESTED STATEMENT	ANSWER
converse of "even $\Rightarrow$ square even"	"if n^2 is even, then n is even"
contrapositive	"if n^2 is odd, then n is odd"
converse of "multiple of $4 \Rightarrow$ even"	false; $n = 2$ is a counterexample

KEY POINT Proving a converse proves a different statement.

To prove $P \Rightarrow Q$, prove $P \Rightarrow Q$ itself or prove its equivalent contrapositive $\neg Q \Rightarrow \neg P$.

Direct proof: start from the hypothesis

THE SHAPE To prove a universal implication

$$\text{for every } x \in D, \quad P(x) \Rightarrow Q(x),$$

use this reliable structure.

NEW NUMBERS The hypothesis may provide new numbers: "a is even" supplies an integer r with $a = 2r$. That r is not a conveniently chosen value.

METHOD

1. Let $x \in D$ be **arbitrary**, and assume $P(x)$.
2. Replace words in the hypothesis by their definitions.
3. Use algebra or facts already known to be true.
4. Show $Q(x)$. Because x was arbitrary, the claim holds for every $x \in D$.

NEVER Never assume $Q(x)$ at the start: that is the conclusion you are trying to prove.

The sum of two even integers is even

CLAIM For every $a, b \in \mathbb{Z}$, if a and b are even, then $a + b$ is even.

PROOF Let $a, b \in \mathbb{Z}$ be arbitrary and assume that both are even. By the definition of even, there are integers r and s such that

$$a = 2r \quad \text{and} \quad b = 2s.$$

Therefore

$$a + b = 2r + 2s = 2(r + s).$$

Since $r + s \in \mathbb{Z}$, the number $a + b$ is 2 times an integer. This is exactly the definition of "$a + b$ is even." ∎

ACTIVITY 2

Write a short direct proof

ACTIVITY 2 PAIRS 12 MIN

Prove the following statement directly:

If a is an even integer and b is an integer, then ab is even.

Use this checklist:

1. Let $a, b \in \mathbb{Z}$ be arbitrary and assume that a is even.
2. Use the definition: write $a = 2k$ for some $k \in \mathbb{Z}$.
3. Rewrite ab in the form $2 \times (\text{an integer})$.

Share: one pair writes their proof on the board; the room checks that the definition of "even" is actually used.

A direct-proof walkthrough

METHOD

1. Open with arbitrary allowed values and the hypothesis, not the conclusion.
2. Substitute the definition into the expression you need to study.
3. Finish by showing that your answer has the form required by the conclusion.

PROOF Let $a, b \in \mathbb{Z}$ be arbitrary, and assume a is even. Since a is even, $a = 2k$ for some $k \in \mathbb{Z}$. Thus

$$ab = (2k)b = 2(kb).$$

Because $k, b \in \mathbb{Z}$, we have $kb \in \mathbb{Z}$. Therefore ab is 2 times an integer, so ab is even. ■

THE TRAP $2(kb)$ only proves "even" after we state that kb is an integer.

When a direct proof is not the clearest route

Sometimes it is hard to work forwards from the hypothesis to the conclusion. For a claim $P \Rightarrow Q$, compare the starting points:

- **direct proof**: assume P, derive Q;
- **contrapositive**: assume $\neg Q$, derive $\neg P$;
- **contradiction**: assume P and $\neg Q$, derive something impossible.

The last route works because the negation of $P \Rightarrow Q$ is "P is true **and** Q is false."

2

Contrapositive: prove the equivalent statement

EQUIVALENCE For each x in a fixed domain D, the statements

$$P(x) \Rightarrow Q(x) \qquad \text{and} \qquad \neg Q(x) \Rightarrow \neg P(x)$$

say the same thing in a different form: they are both true or both false.

The original fails exactly when $P(x)$ is true and $Q(x)$ is false. The contrapositive fails in exactly that same situation.

METHOD To prove a universal implication by contraposition:

1. Let $x \in D$ be arbitrary and assume $\neg Q(x)$.
2. Deduce $\neg P(x)$.
3. Because x was arbitrary, conclude $P(x) \Rightarrow Q(x)$ for every $x \in D$.

WHEN TO USE IT When "not Q" is easier to work with than P. You prove the contrapositive, not the converse.

If n^2 is even, then n is even

CLAIM For every $n \in \mathbb{Z}$, if n^2 is even, then n is even.

CHANGE THE VIEWPOINT Instead of starting from "n^2 is even," prove the contrapositive:

If n is odd, then n^2 is odd.

For integers, "not even" means "odd," and "not odd" means "even."

PROOF Let n be an arbitrary odd integer. By definition, $n = 2k + 1$ for some $k \in \mathbb{Z}$. Hence

$$\begin{aligned} n^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1. \end{aligned}$$

Because $k \in \mathbb{Z}$, we have $k^2 \in \mathbb{Z}$, and hence $j = 2k^2 + 2k \in \mathbb{Z}$. Thus $n^2 = 2j + 1$, so n^2 is odd. This proves the contrapositive; therefore the original claim is true. ∎

Proof by contradiction: an impossible assumption

METHOD To prove a target statement S by contradiction:

1. Assume that S is false; that is, assume $\neg S$.
2. Reason carefully from that assumption.
3. Reach a contradiction: for example, both R and $\neg R$, or a fact known to be impossible.
4. Therefore the assumption $\neg S$ was false, so S is true.

CONDITIONAL TARGETS For a target $P \Rightarrow Q$, assuming its negation means assuming both P and $\neg Q$.

THE CONDITION The contradiction must be a consequence of the assumption, not merely a surprising result.

Set up $\sqrt{2}$

CLAIM $\sqrt{2}$ is irrational.

DEFINITION A real number is **rational** if it can be written as p/q, where $p, q \in \mathbb{Z}$ and $q \neq 0$. It is **irrational** if it is not rational.

Lowest terms: no positive integer greater than 1 divides both numerator and denominator. Any rational number can be written this way.

ASSUME, FOR CONTRADICTION Assume that $\sqrt{2}$ is rational.

1. Choose $p, q \in \mathbb{Z}$ with $q \neq 0$, no common positive divisor greater than 1, and $\sqrt{2} = \dfrac{p}{q}$.
2. Square both sides and use $(\sqrt{2})^2 = 2$:

$$2 = \frac{p^2}{q^2}, \qquad \text{so} \qquad 2q^2 = p^2.$$

3. Since $q^2 \in \mathbb{Z}$, this shows that p^2 is even. By the contrapositive theorem (n^2 even $\Rightarrow n$ even), p is even.

Finish $\sqrt{2}$

PROOF, CONTINUED Since p is even, write $p = 2k$ for some $k \in \mathbb{Z}$. Substitute this into $2q^2 = p^2$:

$$2q^2 = (2k)^2 = 4k^2, \qquad \text{so} \qquad q^2 = 2(k^2).$$

Since $k^2 \in \mathbb{Z}$, the number q^2 is even. Because $q \in \mathbb{Z}$, the same theorem tells us that q is even as well.

Both p and q are divisible by 2. This contradicts the choice of a fraction in lowest terms. Therefore the assumption that $\sqrt{2}$ is rational is false, and $\sqrt{2}$ is irrational. ∎

THE CONTRADICTION "No common positive divisor greater than 1" conflicts with "both p and q are divisible by 2."

Contradiction is how you prove something is *unique*

DIVISION ALGORITHM · THE q AND r ARE UNIQUE

For a and $b > 0$, suppose two pairs worked:

$$a = qb + r, \qquad a = q'b + r', \qquad 0 \le r, r' < b.$$

Subtracting, $b(q - q') = r' - r$, so b divides $r' - r$. The bounds force

$$-b < r' - r < b.$$

The only multiple of b strictly inside is 0. So $r' = r$, and then $b(q - q') = 0$ with $b > 0$ gives $q' = q$. ∎

THE SHAPE TO COPY

To prove **at most one** object qualifies: assume two do, show they are equal.

Divisibility plus a bound on size leaves one option.

When one contradiction is not enough, split into cases

CLAIM If p, q and r are prime, then $p^2 + q^2 \neq r^2$.

1 · BOTH ODD Then $p^2 + q^2$ is even, so r^2 is even and r is even. The only even prime is 2, so $p^2 + q^2 = 4$. But $p, q \geq 3$ gives $p^2 + q^2 \geq 18$. Contradiction. ∎

2 · BOTH EVEN The only even prime is 2, so $p = q = 2$ and $r^2 = 8$. Then r is even, so $r = 2$ and $r^2 = 4 \neq 8$. Contradiction. ∎

3 · ONE OF EACH Say $p = 2$, $q = 2q_1 + 1$. Then r is odd, $r = 2r_1 + 1$, and expanding gives $q_1(q_1 + 1) + 1 = r_1(r_1 + 1)$. Both products are even, so their difference is even — but it equals 1. Contradiction. ∎

There are infinitely many primes

PROOF

Suppose not: the primes are exactly $p_1 < p_2 < \cdots < p_n$, a finite list. Consider

$$N = p_1 p_2 \cdots p_n + 1.$$

N is larger than every p_i, so it is not on the list; being greater than 1 it still has some prime divisor, which must therefore be one of the p_i.

But dividing N by p_i leaves remainder 1, since p_i divides the product. So $p_i \nmid N$ — contradiction. The list can never be complete. $\blacksquare$

THE STEP EVERYBODY GETS WRONG

The proof does *not* claim N is prime.

$2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 + 1 = 30031 = 59 \times 509$, which is composite. The argument only needs that N has **some** prime factor outside the list.

ACTIVITY 3

Choose a proof method before writing

ACTIVITY 3 GROUPS OF THREE 12 MIN

For each claim, choose the proof method that makes the starting condition easiest to use. Then write the first two lines, including the domain and the assumption.

CLAIM	YOUR METHOD
If $a, b \in \mathbb{Z}$ are odd, then $a + b$ is even.	?
For every $n \in \mathbb{Z}$, if n^2 is odd, then n is odd.	?
$\sqrt{2}$ is irrational.	?

Share: each group defends one row. Where two groups picked different methods, both are presented and compared.

Why these methods fit

CLAIM	GOOD WAY TO START
odd $a, b \in \mathbb{Z} \Rightarrow a + b$ even	**Direct:** let $a = 2r + 1$ and $b = 2s + 1$ for $r, s \in \mathbb{Z}$.
n^2 odd $\Rightarrow n$ odd, $n \in \mathbb{Z}$	**Contrapositive:** prove "n even $\Rightarrow n^2$ even." Let $n = 2k$ for $k \in \mathbb{Z}$; then $n^2 = 2(2k^2)$ and $2k^2 \in \mathbb{Z}$.
$\sqrt{2}$ irrational	**Contradiction:** assume it is rational and choose $\sqrt{2} = p/q$ in lowest terms.

METHOD **Choosing a method is part of writing a proof.** Ask: which starting assumption gives me algebra I can use straight away, and what definition will my final line need?

A counterexample can disprove a "for every" claim

NEGATION RULES

$\neg$(for every $x \in D$, $P(x)$) $\iff$ there exists $x \in D$ with $P(x)$ false.

$\neg$(there exists $x \in D$ with $P(x)$) $\iff$ for every $x \in D$, $P(x)$ is false.

COUNTEREXAMPLE To disprove "for every \in D (x) , give an \in D (a).

EXAMPLES

FALSE "FOR EVERY" STATEMENT	COUNTEREXAMPLE
Every prime number is odd.	2 is prime and even.
For every real x, $x^2 \geq x$.	At $x = \frac{1}{2}$, $x^2 = \frac{1}{4} < \frac{1}{2} = x$.

BOTH DIRECTIONS One allowed $a \in D$ with $P(a)$ true proves "there exists $x \in D$ with $P(x)$," but normally does **not** prove a claim about every value. To disprove "there exists," show that no allowed value works.

ACTIVITY 4

Disprove, prove, or explain why one example is not enough

ACTIVITY 4 PAIRS 12 MIN

Decide true or false, with the requested justification. For a counterexample: name the value, confirm it is in the domain, show what fails.

1. For every real x, $x^2 \geq 0$. If true, state the known fact you use.
2. For every integer n, $n^2 + n$ is odd. If false, give a counterexample.
3. For every real x, $(x+1)^2 = x^2 + 1$. If false, give a counterexample.
4. There exists a real x with $x^2 = -1$. Explain why checking a few values is not a disproof; use a fact about **every** real x instead.

Share: one pair gives a counterexample, another explains why checking values cannot settle the last item.

Check "every" or "there exists" first

ITEM	REASONING
1	True. A square of a real number is non-negative: $x^2 \geq 0$ for every $x \in \mathbb{R}$.
2	False. The allowed integer $n = 1$ gives $n^2 + n = 2$, which is even, not odd.
3	False. The allowed real number $x = 1$ gives left side 4 and right side 2.
4	False. For every $x \in \mathbb{R}$, $x^2 \geq 0 > -1$, so no real x can satisfy $x^2 = -1$.

METHOD For item 4, a single failed trial is not enough: the claim says there exists a value. We disprove it by showing that every real value fails.

None of these three proves infinitely many statements at once

THE PROBLEM

Consider the claim: for every $n \geq 1$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

This is not one statement. It is a statement $S(1)$, and $S(2)$, and $S(3)$, ... — infinitely many, one for each n.

WHY CHECKING DOES NOT FINISH

Verifying $S(1), \ldots, S(5)$ proves exactly five of them. That is the "examples are not a proof" rule from this morning, and no amount of further checking escapes it.

WELL-ORDERING PRINCIPLE

Every non-empty set of positive integers has a **least element**.

Obvious-looking, and it is exactly the tool that turns "infinitely many statements" into a single proof by contradiction — on the next slide.

Induction is a proof by contradiction in disguise

PROOF

Suppose $S(1)$ holds, and $S(k)$ always forces $S(k+1)$.

Assume $S(n)$ fails somewhere. Those failures form a non-empty set, so by well-ordering it has a least element m.

$m \neq 1$, since $S(1)$ holds. So $m - 1 \geq 1$.

m is the smallest failure, so $S(m-1)$ is true — and the second hypothesis then forces $S(m)$.

$S(m)$ is true and false. There were no failures. ■

WHAT THIS BUYS YOU Two finite checks replace infinitely many verifications.

TOMORROW Session 6 makes this a template: base case, hypothesis, step.

Four common proof mistakes

RECALL Not (every value has it) means there exists a failure; not (there exists one with it) means every value fails.

PITFALL	FIX
Proving $Q \Rightarrow P$ when the question asks for $P \Rightarrow Q$.	Label the hypothesis and conclusion before you start.
Assuming the conclusion.	Begin with P (direct), $\neg Q$ (contrapositive), or the negated target (contradiction).
Negating "every" or "there exists" incorrectly.	Use the recall above: the quantifier flips and the property is negated.
Giving a few examples as a proof for all values.	Examples test an idea; a proof must cover every case.

EXIT TICKET

Work independently

TASK Complete questions 3 and 4. Then choose **one** full proof from questions 1 and 2.

1 Prove directly: if $m, n \in \mathbb{Z}$ are odd, then $m + n$ is even.

2 Prove by contraposition: for every $n \in \mathbb{Z}$, if n^2 is even, then n is even.

3 Disprove: for every real x, $x^2 + 1 > 2x$.

4 Write the **logical negation** of: "For every real x, $x^2 > x$."

RULE Write the method name and at least one sentence explaining why your proof or counterexample works. For a counterexample, state why the chosen value belongs to the stated domain.

WRAP-UP

What today was about

- → State the **domain** and quantifier first.
- → A converse is a different statement; a contrapositive is true in exactly the same cases.
- → **Direct** starts from the hypothesis; **contrapositive** from "not the conclusion"; **contradiction** from the negated target.
- → One **counterexample** disproves a claim about every allowed value.
- → Contradiction also proves **uniqueness**, and handles **exhaustive cases**.

Problem Set 5

Eight statements to prove or disprove. Name the method at the top of each solution and explain each key step.

Next session: **mathematical induction** — the template built on the well-ordering argument we just saw.

implication · converse · contrapositive · negation · quantifier · contradiction · counterexample · uniqueness · well-ordering